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Tape Security With DFSMSrmm

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Want to know more about protecting tape data on z/OS using RACF and DFSMSrmm? Come to this session to learn more about how tape data can be protected and about options available in DFSMSrmm to help manage RACF profiles. Find out how tape volumes are identified, protected, and managed and how RACF security options for tape provide you flexibility in setting up your tape security. This session includes a preview of changes in tape data set authorization with z/OS V1R8.

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Agenda

- **Overview of Tape Security**
 - What you get today
 - Planned z/OS 1.8 Enhancements
- **Details of Tape Authorization**
 - TAPEVOL and TAPEDSN
 - So, What Options Will I have?
- **z/OS 1.8 Implementation**
- **Additional Considerations**



Overview

- **DFSMSdfp OPEN**

- SAF Control points for tape security
 - RACROUTE with AUTH/DEFINE

- **RACF**

- Provides Choices for Tape Security
- Takes All Security Decisions based on Rules and Profiles

- **DFSMSrmm**

- Provides Options for Managing RACF Profiles

SAF = Systems Authorization Facility – Routes to installed security product (RACF)

Existing Tape Security Choices

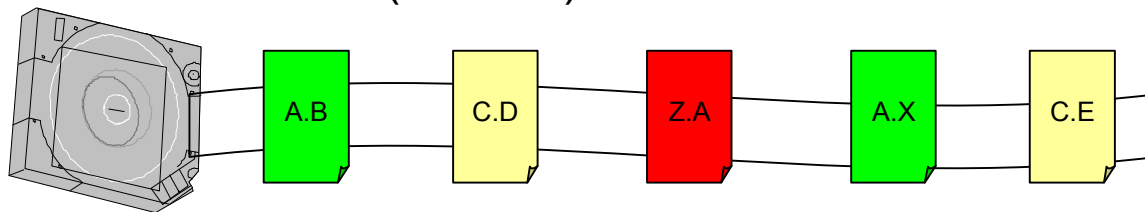
■ RACF

– SETROPTS TAPEDSN

- DFSMS Uses DATASET class for LABEL=SL/AL
- RACF checks DATASET + TAPEVOL profiles

– TAPEVOL Class

- DFSMS uses TAPEVOL class
- RACF checks TAPEVOL profiles
- FACILITY class ICHBLP checked for LABEL=BLP
- TAPEVOL optionally includes a Tape Volume Table Of Contents (TVTOC)



VOL001

DFSMSdfp OPEN

- **Processing Depends on SETROPTS**
 - SETROPTS NOTAPEDSN NOCLASSACT(TAPEVOL)
 - No Checking
 - SETROPTS TAPEDSN
 - RACROUTE in DATASET class for SL/AL
 - SETROPTS CLASSACT(TAPEVOL)
 - RACROUTE in TAPEVOL class
- **Processing Depends on LABEL type**
 - BLP/NL Always in TAPEVOL class

DFSMSrmm

- **Manages TAPEVOL and Discrete DATASET Profiles**
 - Exact Processing Depends on RACF SETROPTS
 - TAPEDSN Discrete DATASET profiles cleaned up
 - TAPEVOL TAPEVOL Created and Cleaned up
 - TAPEDSN+TAPEVOL TAPEVOL+TVTOC for SCRATCH
 - OPTION TPRACF Controls RMM Processing
 - NONE No Management
 - PREDEFINED & AUTOMATIC
 - TAPEVOL created and Cleaned Up
 - DATASET Profiles Cleaned Up
 - TAPEVOL Access List Maintained
 - > Volume OWNER added
 - > Access List from Volume Record

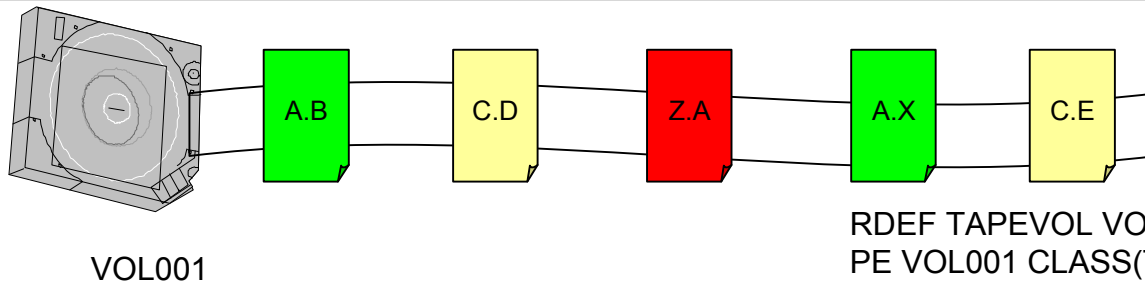
RMM Never Authorizes Access to Tape Data Sets

Example no TVTOC

```
//READ JOB (acct),USER=USERA
```

```
//TAPE DD DISP=SHR,VOL=SER=VOL001,LABEL=(1,SL),DSN=A.B
```

SETROPTS	DFSMSdfp OPEN	RACF PROFILE
TAPEDSN	RACROUTE CLASS=DATASET,DSN=A.B,DSTYPE=T,LABEL=SL	DATASET 'A.*'
TAPEVOL	RACROUTE CLASS=TAPEVOL,LABEL=SL	TAPEVOL VOL001
TAPEVOL + TAPEDSN	RACROUTE CLASS=DATASET,DSN=A.B,DSTYPE=T,LABEL=SL	TAPEVOL VOL001



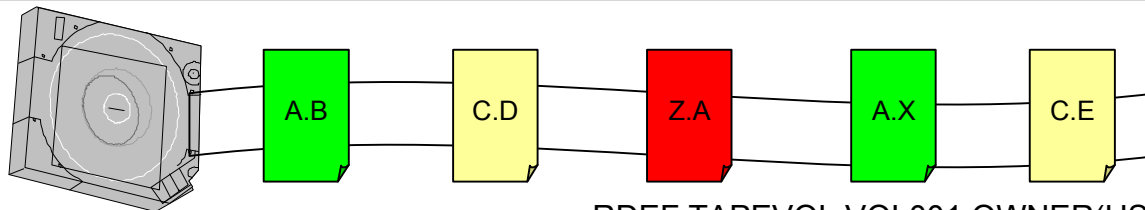
```
RDEF TAPEVOL VOL001 OWNER(USERA) UAC(NONE)
PE VOL001 CLASS(TAPEVOL) ID(USERB) ACC(READ)
ADDSD 'A.*' OWNER(USERA) UAC(NONE)
PE 'A.*' ID(USERC) ACC(READ)
```

Example with TVTOC

```
//READ JOB (acct),USER=USERC
```

```
//TAPE DD DISP=SHR,VOL=SER=VOL001,LABEL=(1,SL),DSN=A.B
```

SETROPTS	DFSMSdfp OPEN	RACF PROFILE
TAPEDSN	RACROUTE CLASS=DATASET,DSN=A.B,DSTYPE=T,LABEL=SL	DATASET 'A.*'
TAPEVOL	RACROUTE CLASS=TAPEVOL,LABEL=SL	TAPEVOL VOL001
TAPEVOL + TAPEDSN	RACROUTE CLASS=DATASET,DSN=A.B,DSTYPE=T,LABEL=SL	DATASET 'A.*'



VOL001

```
RDEF TAPEVOL VOL001 OWNER(USERA) UAC(NONE) TVTOC
```

```
PE VOL001 CLASS(TAPEVOL) ID(USERB) ACC(READ)
```

```
ADDSD 'A.*' OWNER(USERA) UAC(NONE)
```

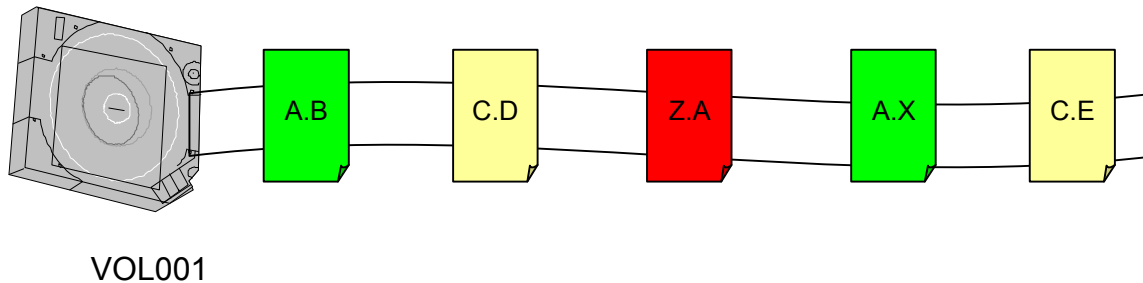
```
PE 'A.*' ID(USERC) ACC(READ)
```

```
ADDSD 'A.B' TAPE VOLUME(VOL001) FILESEQ(1) UNIT(TAPE) SETONLY
```

Example with TVTOC ...

- **TVTOC Example is incomplete without TVTOC Entries**
 - Add remaining file information

```
ALTDSD 'A.*' RETPD(5)
ADDSD 'C.*' OWNER(USERA) UAC(NONE) RETPD(6)
PE 'C.*' ID(USERC) ACC(READ)
ADDSD 'Z.*' OWNER(USERA) UAC(NONE) RETPD(6)
PE 'Z.*' ID(USERC) ACC(READ)
ADDSD 'C.D' TAPE VOLUME(VOL001) FILESEQ(2) UNIT(TAPE) SETONLY
ADDSD 'Z.A' TAPE VOLUME(VOL001) FILESEQ(3) UNIT(TAPE) SETONLY
ADDSD 'A.X' TAPE VOLUME(VOL001) FILESEQ(4) UNIT(TAPE) SETONLY
ADDSD 'C.E' TAPE VOLUME(VOL001) FILESEQ(5) UNIT(TAPE) SETONLY
```



TVTOC Provides

- **Best option for tape data set security**
 - Automatic protection for data written to scratch pool tapes
- **RETPD checking for overwritten files**
- **44 Character Data Set Name Checking**
- **Discrete or Generic DATASET Profile Usage**
- **Useful Information when you have No Tape Manager**
- **Restrictions**
 - Profile size
 - Numbers of Data Sets and volumes
 - Tape Manager cannot remove a single volume from a volume set
- **Problems**
 - Numbers of TAPEVOL profiles
 - Management of profiles

Problems

- **Tape Volume Labels contain only last 17 characters of data set name**
 - Tape Manager provides full 44 character checking
- **Authorization to One Data Set Allows Access to All others**
 - Application can reposition the tape to access any data
- **TVTOC Limitations**
- **No SAF call unless RACF SETROPTS enable one method of Tape Security**

Requirements

Tape data set authorization needs to exploit existing DATASET profiles just as for DASD data sets.

Overcome the limitations of RACF TVTOC support.

Support Common Authorization for multi-file Volumes

Brief Description	Source
Provide tape data set authorization independent of the TAPEVOL class and TAPEDSN option enabling generic DATASET profiles to be used for both DASD and Tape data sets	IBM Storage Symposium DSSMSD03001
Allow DFSMSrmm to use the ERASE field in RACF generic data set profiles to determine if tape data sets should be erased when deleted or the tape is returned to SCRATCH status	MR020205712

z/OS 1.8 Preview

New Tape Data Set Authorization Options Provided by DFSMS

- **DFSMSdfp**

- Option to Always Call SAF in DATASET class
- Erase-on-Scratch Setting Awareness
- Additional SAF Calls for File 1 and Overwritten Files
- SAF Return Code Options
- Exploitation by Tape Manager Required for Some Options

- **DFSMSrmm Exploits the New Options**

- ERASE Release Action set
- File 1 Data Set Name Provided to Enable Additional SAF Calls
- SAF Return Code Altered for Ignored Volumes

New DEVSUP options

■ TAPEAUTHDSN

Use this new keyword to enable tape data set authorization checks in the DATASET class as for DASD.

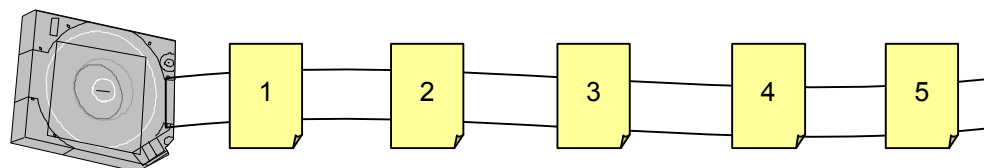
The data set name specified in the allocation/JCL is used by the system to check your authorization to read or write the specified file.

TAPEAUTHDSN = { YES | NO }

TAPEAUTHF1 = { YES | NO }

TAPEAUTHRC4 = { ALLOW | FAIL }

TAPEAUTHRC8 = { FAIL | WARN }



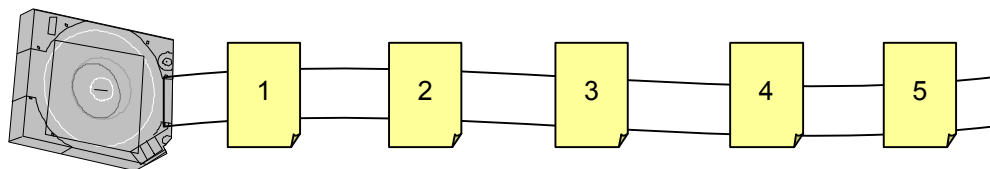
New DEVSUP options

■ TAPEAUTHF1

Use this new keyword to enable tape data set authorization checks in the DATASET class for the first file on the same tape volume when any other file on the same tape volume is opened.

TAPEAUTHDSN = { YES | NO }
TAPEAUTHF1 = { YES | NO }
TAPEAUTHRC4 = { ALLOW | FAIL }
TAPEAUTHRC8 = { FAIL | WARN }

Note: When neither TAPEAUTHDSN nor SETROPTS TAPEDSN is in use, TAPEAUTHF1 support is not provided.

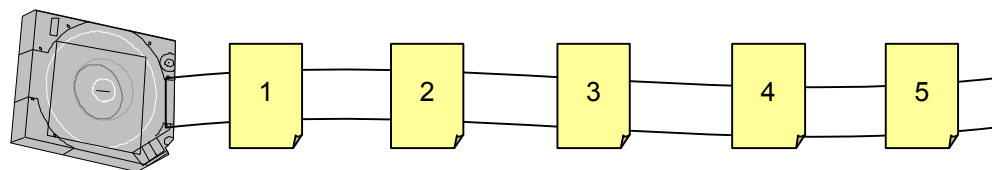


New DEVSUP options

■ TAPEAUTHRC4

Use this new keyword to control PROTECTALL processing for tape data sets. The default is FAIL; data sets which are not protected by a security profile cannot be accessed.

TAPEAUTHDSN = { YES | NO }
TAPEAUTHF1 = { YES | NO }
TAPEAUTHRC4 = { ALLOW | FAIL }
TAPEAUTHRC8 = { FAIL | WARN }

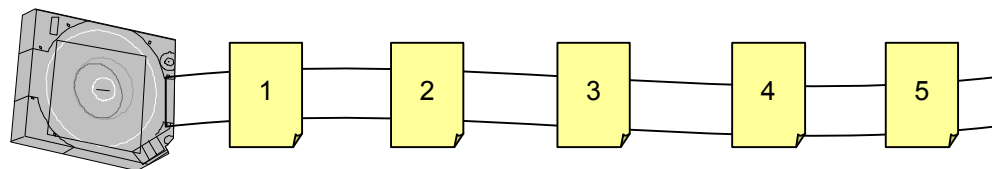


New DEVSUP options

■ TAPEAUTHRC8

Use this new keyword to provide a managed and controlled implementation of tape data set authorization checks in the DATASET class. When the value is WARN; data sets which normally could not be accessed, are allowed to be and a warning message is issued.

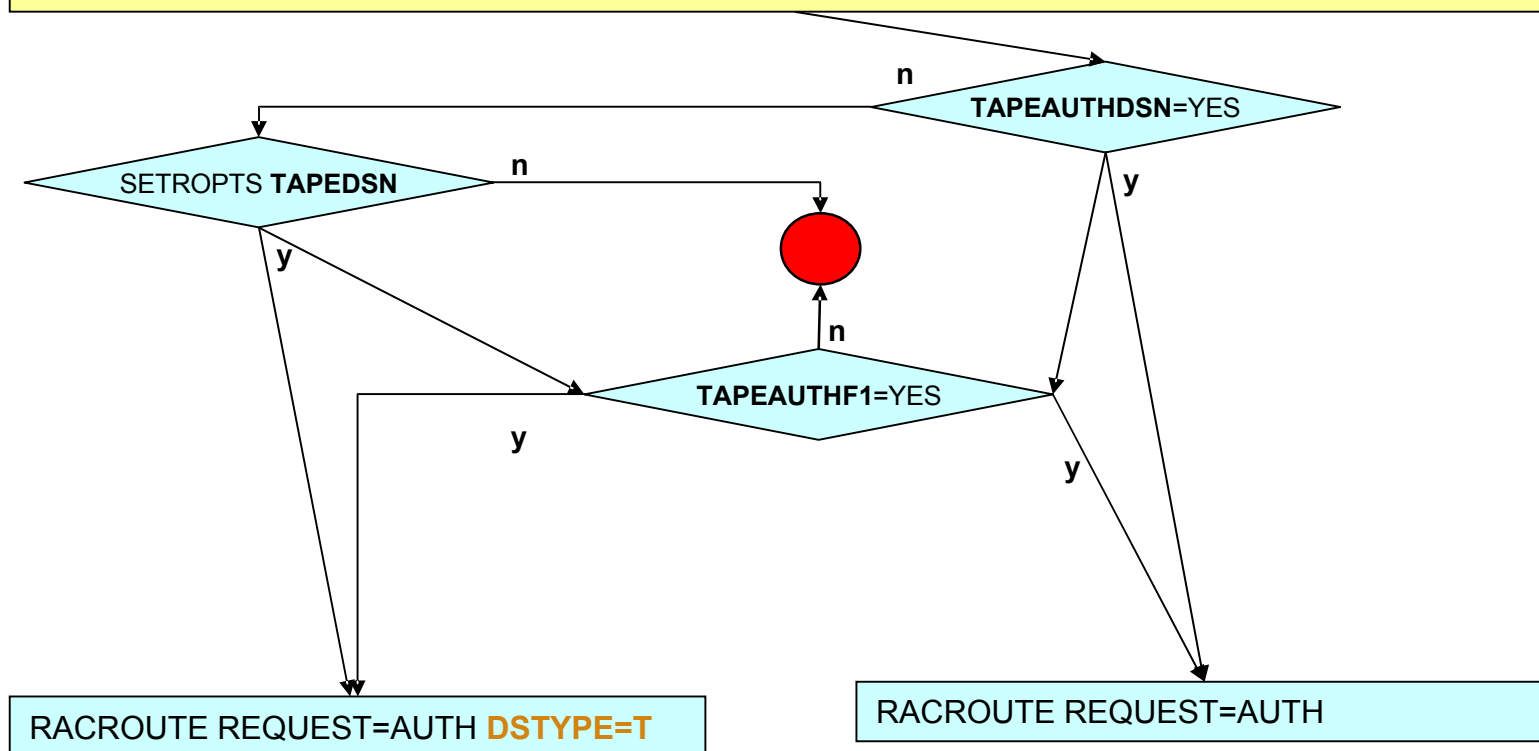
TAPEAUTHDSN = { YES | NO }
TAPEAUTHF1 = { YES | NO }
TAPEAUTHRC4 = { ALLOW | FAIL }
TAPEAUTHRC8 = { FAIL | WARN }



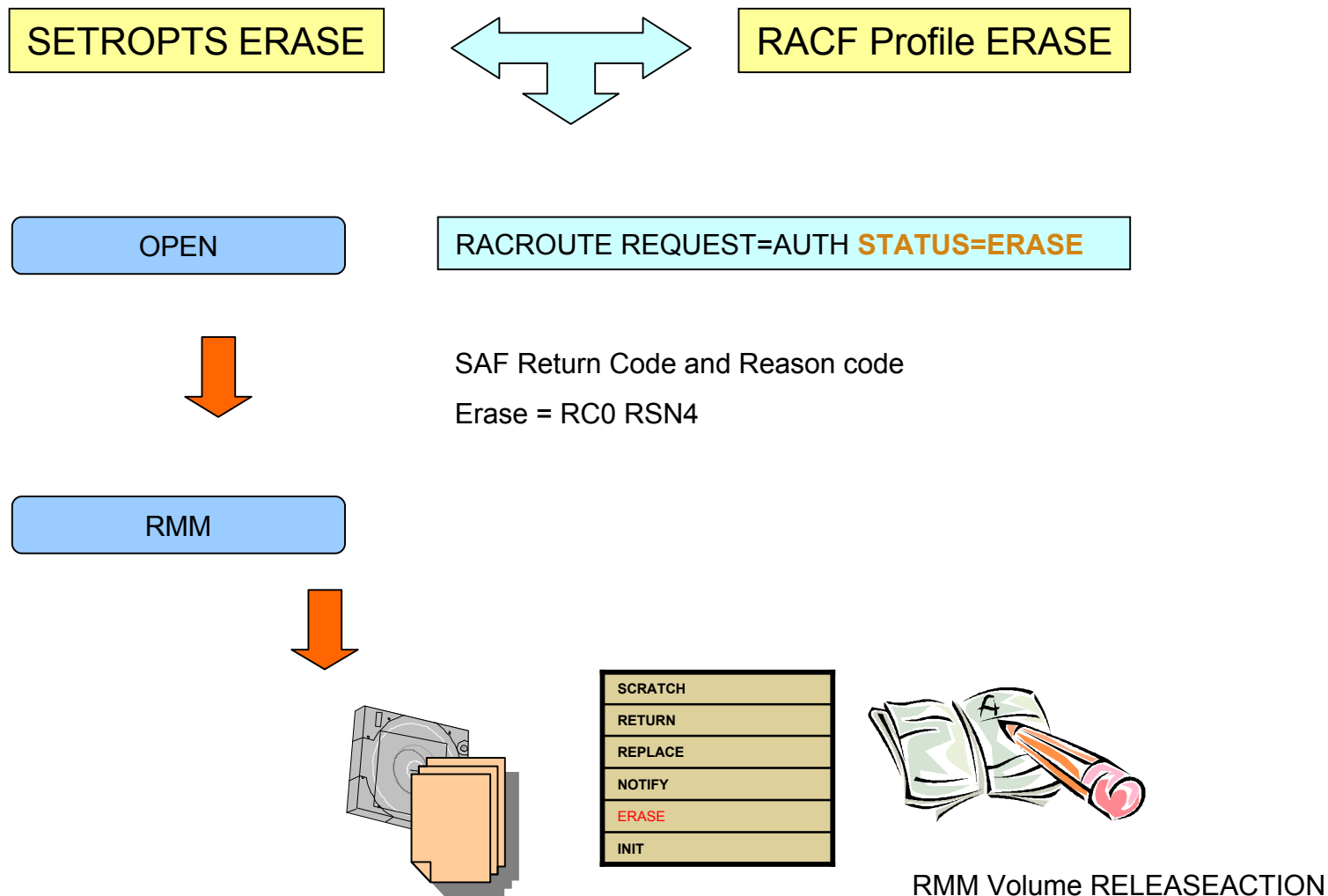
DSTYPE=T ?

DSTYPE=T indicates to RACF that the check is for data set on tape volume and that special RACF tape data set and tape volume processing is to be performed.

Without DSTYPE=T RACF authorization checking considers only profiles in the DATASET class.



RACF ERASE on SCRATCH support



DFSMSrmm Exploitation

- **Manages TAPEVOL and Discrete DATASET Profiles**
 - OPTION TPRACF Controls RMM Processing
 - CLEANUP Clean up Discrete Profiles
- **ERASE Release Action Set**
- **TAPEAUTHF1=YES**
 - File 1 Data Set Name and SEQ Passed to OPEN
 - When Subsequent file opened
 - When existing first File Overwritten
- **Ignored Volumes**
 - STGADMIN.EDG.IGNORE.TAPE.
 - Enables Support and SAF RC changed to RC0
 - Authorizes access to data set not defined to RACF

RMM Ignored Volumes

//READ JOB (acct),USER=USERC

//TAPE DD DISP=SHR,VOL=SER=VOL001,LABEL=(1,SL),DSN='@-/.A.B'

```
RDEF FACILITY STGADMIN.EDG.IGNORE.TAPE.NORMM.VOL001
      UAC (NONE)
```

```
PE STGADMIN.EDG.IGNORE.TAPE.NORMM.VOL001 ID (A) AC (R)
```

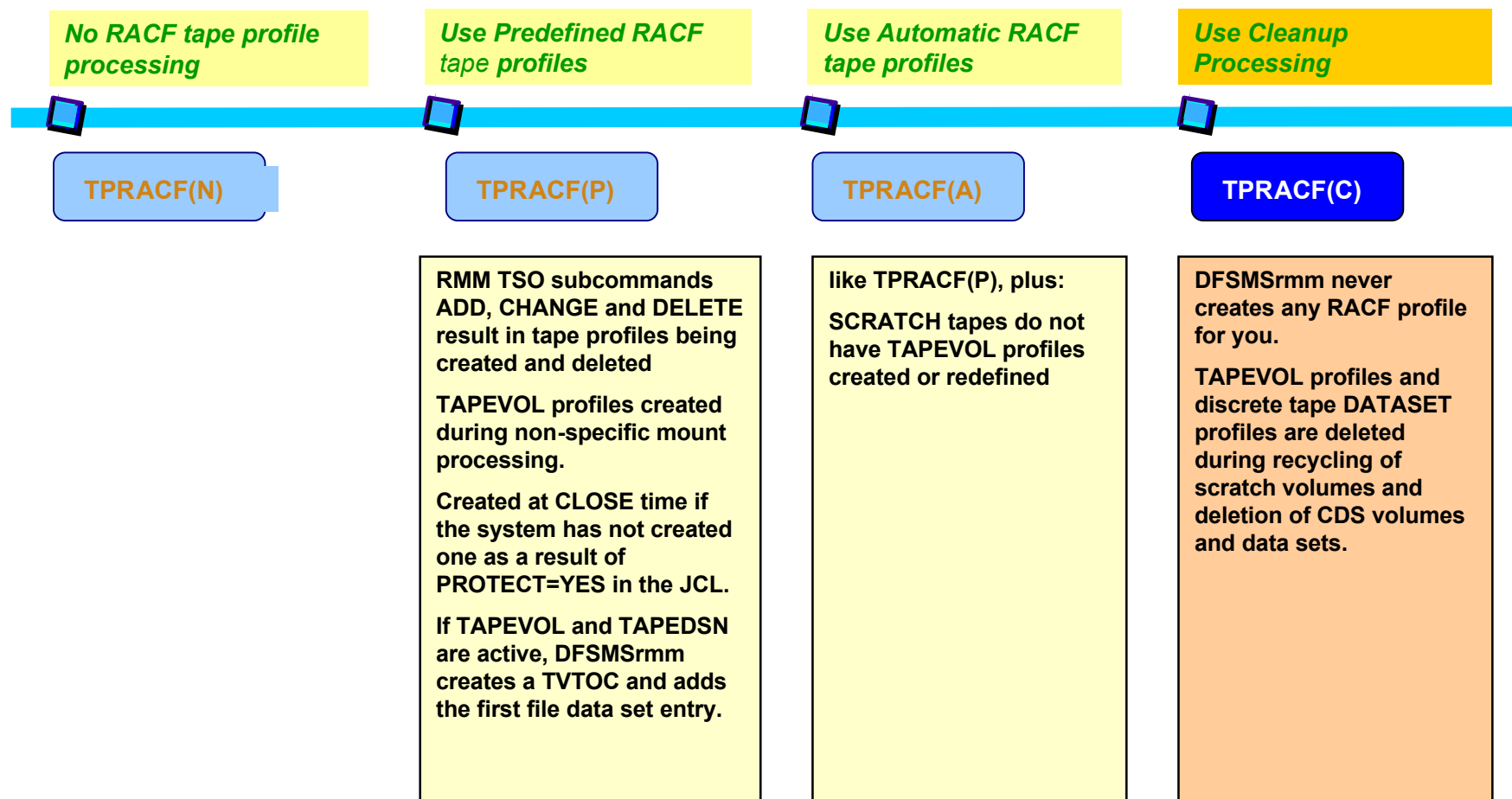
NO TAPEVOL
NO DATASET Profile

OPEN issues RACROUTE DATASET/TAPEVOL
SAF RC4 (Not Protected)

RMM issues RACROUTE FACILITY STGADMIN.EDG.IGNORE.TAPE.NORMM.VOL001
SAF RC0 (Access Allowed)

RMM Sets OPENs SAF to RC0
Allowing access

Tape Security – Parmlib Option TPRACF



Implementation

Target System Configuration

■ **DEVSUPxx**

```
TAPEAUTHDSN=YES  
TAPEAUTHF1=YES  
TAPEAUTHRC4=FAIL  
TAPEAUTHRC8=FAIL
```

■ **EDGRMMxx**

```
OPTION TPRACF(NONE)
```

■ **RACF**

```
SETROPTS NOTAPEDSN NOCLASSACT(TAPEVOL)
```

But You need some Buddies to Help You Get There . . .

Your Buddies

During Implementation/Migration Only

- **DEVSUPxx**

TAPEAUTHRC4=ALLOW
TAPEAUTHRC8=WARN

- **EDGRMMxx**

OPTION TPRACF(CLEANUP)

TAPEAUTHRC4 Allows access to unprotected tape data sets until you can add missing DATASET profiles.

TAPEAUTHRC8 Allows access while existing DATASET profiles are adjusted to support tape authorization

CLEANUP aids the deletion of existing TAPEVOL profiles and discrete DATASET profiles as tape volumes are recycled to scratch status.

Do not use this DFSMSrmm option unless you have successfully implemented TAPEAUTHDSN

And There is More . . .

Other Considerations

■ DFMSHsm

```
SETSYS  
TAPESECURITY(RACF/RACFINCLUDE)  
ADDSD 'mprefix.**'  
ADDSD 'bprefix.**'  
ADDSD 'authid.**'  
ADDSD 'bprefix.DMP.**'  
ADDSD 'outputdatasetprefix.**'
```

TAPEAUTHDSN avoids use of HSM
TAPEVOL profiles.
Create DATASET Profiles

Change the HSM Setting after
implementation.

■ IEHINITT

IEHINITT issues RACROUTE in TAPEVOL class for existing volumes

When using TAPEAUTHDSN or TAPEDSN you can use one or more
generic TAPEVOL for IEHINITT

Prefer EDGINERS for tape labelling

Other Considerations . . .

■ **Non-IBM Security Products**

- DFSMSdfp uses the SAF interface
 - Any SAF compliant security product can support the new options
- RACROUTEs for TAPEAUTHDSN are not 'Tape'

■ **RACF**

- No Changes were made to RACF
- Existing DATASET class Profiles and RACROUTE capabilities are exploited

Recommendations



1. **If Using TAPEDSN and TAPEVOL**
 - **Replace with TAPEAUTHDSN and TAPEAUTHF1**
2. **If Using TAPEDSN;**
 - **Use TAPEAUTHF1 to enable Common Authorization**
3. **Use TAPEAUTHRC8 to ease implementation**
4. **Convert RMM SECCLS ERASE security class settings to RACF DATASET Profiles**

Summary

■ z/OS 1.8 Tape Data Set Authorization Choices

- Planned GA September, 2006
- Simplifies Tape Security Implementation
- Supports all Tape Volumes
 - Including Foreign Tapes
- Enables
 - Common Authorization at Volume Level
 - Choice over implementation
 - Automates RACF ERASE on SCRATCH
 - Single Authorization Scheme for all Data Sets

